林仲豪

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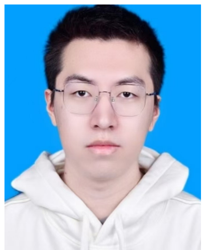

教育经历

北京理工大学　控制工程　硕士研究生　　2024 年 9 月 – 至今

均分：92.40 / 100；导师：曾宪琳教授；

研究方向：时变优化、在线优化、分布式优化及其在目标定位与跟踪问题中的应用。

北京理工大学　自动化　工学学士　　2020 年 9 月 – 2024 年 6 月

均分：91.52 / 100，专业前 3%；英语成绩：四级：624，六级：580。

论文发表（一作）

1. **Z. Lin**, J. Hou, and X. Zeng, "Optimal Prediction-Correction Algorithm Using Sparse Linear Extrapolation for Time-Varying Optimization," *IEEE Transactions on Signal Processing*, vol. 74, pp. 2564-2578, 2026.
2. **Z. Lin**, X. Zeng, J. Hou, J. Sun, and J. Chen, "Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization," *Journal of Systems Science and Complexity*, vol. 39, no. 2, pp. 483–510, 2026.
3. **Z. Lin**, J. Hou, and X. Zeng, "Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization," *IEEE Conference on Decision and Control (CDC 2026)*, **已录用**。
4. **Z. Lin**, "Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking," 拟投稿至 *IEEE Signal Processing Letters*。

研究经历

时变优化的预测-校正方法（硕士课题）　　2024 年 9 月 – 至今

1. 提出面向强凸时变优化问题的**稀疏线性外推预测-校正（SE-PC）**方法，在可处理约束下通过 ℓ_1 范数最小化设计预测器系数。该方法在达到高阶跟踪精度的同时，显著减少了收敛所需的梯度下降校正步数。相关成果**已发表**于信号处理领域旗舰期刊 *IEEE Transactions on Signal Processing (TSP)*。
2. 提出面向线性约束时变凸优化问题的**可调记忆原始-对偶预测-校正（PD-PC）**方法，将可调记忆外推预测与原始-对偶梯度校正相结合，在时变线性约束下实现了二阶渐近跟踪精度。相关成果**已发表**于卓越期刊 *Journal of Systems Science and Complexity (JSSC, JCR Q1)*。
3. 提出面向连续时间时变凸优化问题的**无 Hessian 逆矩阵预测-校正流**，通过引入辅助矩阵动力学在线构造 Hessian 逆矩阵的替代估计，从而避免显式矩阵求逆。相关成果**已录用**于控制领域顶会 *IEEE Conference on Decision and Control (CDC 2026)*。
4. 将 TSP 工作中的线性外推预测-校正方法拓展至基于到达时间（TOA）的目标定位与跟踪问题。针对 TOA 最小二乘目标函数**仅局部强凸、整体非凸**的特点，证明了外推预测点与校正迭代点均保持在当前目标函数的局部强凸区域内；进一步给出了显式跟踪误差界，表明算法的跟踪误差以 Q-线性速度收敛至噪声相关邻域，且该邻域在无噪声情形下退化为零。相关成果**拟投稿**至信号处理领域快报 *IEEE Signal Processing Letters (SPL)*。

主要荣誉、奖励与学术服务

1. **参与审稿：*IEEE ICCA 2026*、*CCC 2026***　　2026
2. **北京理工大学研究生特等奖学金** (约前 5%)　　2024
3. **北京理工大学优秀毕业生**　　2024
4. **三星奖学金**（排名：3 / 200+）　　2023
5. **全国大学生数学竞赛一等奖**（非数学类，共两次）　　2021, 2022
6. **北京理工大学学业一等奖学金**（专业前 5%，共四次）　　2021–2024
7. **北京理工大学新生数学竞赛一等奖**　　2020

附录: 论文摘要

1. 已发表（长文） **Z. Lin**, J. Hou, and X. Zeng, "Optimal Prediction-Correction Algorithm Using Sparse Linear Extrapolation for Time-Varying Optimization," *IEEE Transactions on Signal Processing*, vol. 74, pp. 2564-2578, 2026.

英文摘要：This paper introduces an optimal prediction-correction algorithm leveraging sparse linear extrapolation for strongly convex, unconstrained time-varying optimization problems, which are prevalent in dynamic systems and online learning. The proposed method constructs the prediction phase as a sparse linear combination of past iterates, with extrapolation coefficients derived by solving an ℓ_1-norm minimization problem under tractable constraints. By promoting sparsity in the predictor, the algorithm reduces the frequency of correction steps and the associated computational cost of gradient evaluations. We establish the existence of an ℓ_1-optimal sparse predictor and derive closed-form solutions for second- and third-order tracking accuracy cases. Theoretical analysis confirms that the method achieves state-of-the-art tracking accuracy with improved computational efficiency compared to existing prediction-correction approaches. Numerical experiments validate the theoretical results, demonstrating the advantages of the proposed algorithm in reducing computational overhead while maintaining high accuracy.

2. 已发表（长文）: **Z. Lin**, X. Zeng, J. Hou, J. Sun, and J. Chen, "Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization," *Journal of Systems Science and Complexity*, vol. 39, no. 2, pp. 483–510, 2026.

英文摘要：This paper presents a primal-dual prediction-correction (PD-PC) method for solving linearly constrained time-varying convex optimization problems, which frequently arise in control, signal processing, and online learning applications. The proposed method establishes a novel integration of primal-dual gradient dynamics with a discrete-time prediction-correction structure, specifically designed for problems with time-dependent linear constraints. A tunable memory parameter is introduced in the prediction phase to perform linear extrapolation using past iterates, enabling a flexible trade-off between the amount of historical information stored and the computational cost of correction. In the correction phase, primal and dual variables are updated via gradient descent-ascent iterations, thus maintaining the computational efficiency of a first-order method without requiring Hessian or high-order derivative computations. Theoretical analysis shows that the method achieves $\mathcal{O}(h^2)$ asymptotic tracking accuracy for both primal and dual variables, matching the state-of-the-art performance among first-order methods even in unconstrained settings. Numerical experiments on problems with both time-invariant and time-varying constraints validate the theoretical findings and demonstrate the method's effectiveness.

3. 已录用，待发表: **Z. Lin**, J. Hou, and X. Zeng, "Hessian-Inverse-Free Prediction-Correction Method for Time-Varying Convex Optimization," *IEEE Conference on Decision and Control (CDC 2026)*.

英文摘要：In this paper, we propose a Hessian-inverse-free prediction-correction method for time-varying convex optimization. The proposed method embeds an auxiliary H flow into a prediction-correction x flow, where the H flow generates an online surrogate of the Hessian inverse through prediction, correction, and anchoring terms, thereby avoiding explicit matrix inversion. Under uniform strong convexity, the coupled x–H flow guarantees convergence of the gradient norm to zero and asymptotic tracking of the time-varying optimal trajectory. In the merely convex case, the proposed method remains well-posed and yields a tunable asymptotic bound on the gradient norm. A numerical example illustrates the efficacy of the proposed method.

4. 草稿: **Z. Lin**, "Linear-Extrapolation Prediction-Correction for TOA-Based Target Tracking,"

英文摘要 (草稿)：This letter studies target tracking from sequential time-of-arrival (TOA) measurements. The problem is formulated as a time-varying TOA least-squares problem, whose objective is generally nonconvex and locally strongly convex only near its least-squares estimate. Motivated by polynomial and spline trajectory models used in autonomous systems, we propose a linear-extrapolation prediction-correction approach for TOA-based tracking. The method predicts the next target position from past iterates and then applies a few gradient correction steps to the current TOA loss. We prove that, under standard local TOA regularity conditions, both the extrapolated predictions and the corrected iterates remain inside the local strong-convexity region of the current TOA least-squares objective, leading to an explicit tracking bound with respect to the true target trajectory. The bound further shows that the tracking error converges Q-linearly to a noise-dependent neighborhood, which reduces to zero in the noiseless case.